1) a)

$y = \frac{2}{3}x^{\frac{3}{2}}$ so $\frac{dy}{dx} = x^{\frac{1}{2}}$

$s = \int_8^{35} \sqrt{1 + \left(x^{\frac{1}{2}}\right)^2}\, dx = \int_8^{35} (1+x)^{\frac{1}{2}} dx$

$= \left[\frac{2}{3}(1+x)^{\frac{3}{2}}\right]_8^{35} = 144 - 18 = 126$ units

1) b)

$y = \frac{4}{3}x^{\frac{3}{2}}$ so $\frac{dy}{dx} = 2x^{\frac{1}{2}}$

$s = \int_1^4 \sqrt{1 + \left(2x^{\frac{1}{2}}\right)^2}\, dx = \int_1^4 (1+4x)^{\frac{1}{2}} dx$

$= \left[\frac{1}{6}(1+4x)^{\frac{3}{2}}\right]_1^4 = \frac{17}{6}\sqrt{17} - \frac{5}{6}\sqrt{5}$ units

2)

$y = \cosh x$ so $\frac{dy}{dx} = \sinh x$

$s = \int_{\ln 2}^{\ln 3} \sqrt{1 + (\sinh x)^2} dx = \int_{\ln 2}^{\ln 3} \sqrt{\cosh^2 x}\, dx = \int_{\ln 2}^{\ln 3} \cosh x\, dx$

$= [\sinh x]_{\ln 2}^{\ln 3} = \frac{4}{3} - \frac{3}{4} = \frac{7}{12}$

3) a)

$\int \sec x\, dx = \int \sec x \times \frac{\sec x + \tan x}{\sec x + \tan x} dx$

$= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx$

Since $\frac{d}{dx}(\sec x + \tan x) = \sec x \tan x + \sec^2 x$

Then

$\int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx = \ln|\sec x + \tan x|$ as req.

3) b)

$y = \ln(\cos x)$ so $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{-\sin x}{\cos x} = -\tan x$

$s = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \sqrt{1 + (-\tan x)^2}\mathrm{d}x = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \sqrt{\sec^2 x}\,\mathrm{d}x = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \sec x\,\mathrm{d}x$

$= [\ln|\sec x + \tan x|]_{\frac{\pi}{4}}^{\frac{\pi}{3}} = \ln\left(2+\sqrt{3}\right) - \ln\left(1+\sqrt{2}\right) = \ln\left(\frac{2+\sqrt{3}}{1+\sqrt{2}}\right)$ units

4)

$y = \frac{1}{2}x^2$ so $\frac{\mathrm{d}y}{\mathrm{d}x} = x$

$s = \int_0^2 \sqrt{1 + x^2}\mathrm{d}x$

Let $x = \sinh u$ so that $\frac{\mathrm{d}x}{\mathrm{d}u} = \cosh u$

When $x = 0$, $u = \operatorname{arsinh}(0) = 0$, when $x = \ln 2$, $u = \operatorname{arsinh}(2) = \ln\left(2+\sqrt{5}\right)$

$\int_0^2 \sqrt{1 + x^2}\mathrm{d}x = \int_0^{\ln\left(2+\sqrt{5}\right)} \cosh u \sqrt{1 + \sinh^2 u}\,\mathrm{d}u$

$= \int_0^{\ln\left(2+\sqrt{5}\right)} \cosh^2 u\,\mathrm{d}u$

Since $\cosh 2A = 2\cosh^2 A - 1$ then $\cosh^2 A = \frac{1}{2} + \frac{1}{2}\cosh 2A$ so that

$s = \int_0^{\ln\left(2+\sqrt{5}\right)} \cosh^2 u\,\mathrm{d}u = \int_0^{\ln\left(2+\sqrt{5}\right)} \left(\frac{1}{2} + \frac{1}{2}\cosh 2u\right)\mathrm{d}u$

$= \left[\frac{1}{2}u + \frac{1}{4}\sinh 2u\right]_0^{\ln\left(2+\sqrt{5}\right)} = \left(\frac{1}{2}\ln\left(2+\sqrt{5}\right) + \sqrt{5}\right) - (0+0)$

$= \left(\sqrt{5} + \frac{1}{2}\ln\left(2+\sqrt{5}\right)\right)$ units

5)

$x = 2\cosh\left(\frac{y}{2}\right)$ so $\frac{\mathrm{d}x}{\mathrm{d}y} = \sinh\left(\frac{y}{2}\right)$

$s = \int_0^{\ln 4} \sqrt{1 + \left(\sinh\left(\frac{y}{2}\right)\right)^2}\,\mathrm{d}y = \int_0^{\ln 4} \sqrt{\cosh^2\left(\frac{y}{2}\right)}\,\mathrm{d}y = \int_0^{\ln 4} \cosh\left(\frac{y}{2}\right)\mathrm{d}y$

$= \left[2\sinh\left(\frac{y}{2}\right)\right]_0^{\ln 4} = \left(\frac{3}{2}\right) - (0) = \frac{3}{2}$ units

6) a)

The curve goes through $B(0, 15)$:

$15 = k\cosh\frac{0}{k} - p \Rightarrow 15 = k - p$ as required.

6) b)

$15 = k - p \Rightarrow p = k - 15$

The curve goes through $(70, 40)$:

$40 = k\cosh\frac{70}{k} - p$

$\Rightarrow 40 = k\cosh\frac{70}{k} - (k - 15)$

$\Rightarrow 0 = k\cosh\frac{70}{k} - k - 25$

Let $\mathrm{f}(k) = k\cosh\frac{70}{k} - k - 25$

$\mathrm{f}(101.905) = 101.905\cosh\left(\frac{70}{101.905}\right) - 101.905 - 25 = 0.00234\ldots > 0$

$\mathrm{f}(101.915) = 101.915\cosh\left(\frac{70}{101.915}\right) - 101.915 - 25 = -0.000295\ldots < 0$

As there is a change of sign and $\mathrm{f}(k)$ is continuous between the two values of k, we must have that $k = 101.91$ (2 DP)

6) c)

$y = 101.91\cosh\frac{x}{101.91} - 86.91$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \sinh\frac{x}{101.91}$

$s = \int_{-70}^{70}\sqrt{1 + \left(\sinh\frac{x}{101.91}\right)^2}\,\mathrm{d}x$

$= \int_{-70}^{70}\sqrt{1 + \sinh^2\left(\frac{x}{101.91}\right)}\,\mathrm{d}x$

$= \int_{-70}^{70}\sqrt{\cosh^2\left(\frac{x}{101.91}\right)}\,\mathrm{d}x$

$= \int_{-70}^{70}\cosh\left(\frac{x}{101.91}\right)\mathrm{d}x$

$= \left[101.91\sinh\left(\frac{x}{101.91}\right)\right]_{-70}^{70}$

$= \left(101.91\sinh\left(\frac{70}{101.91}\right)\right) - \left(101.91\sinh\left(\frac{-70}{101.91}\right)\right)$

$= 75.635\ldots - (-75.635\ldots)$

$= 151.271\ldots$

$= 151$ metres to the nearest metre.

7) a)

The minimum point of the curve will be when $x = 250$, and the height will be 60 metres. At this point $\cosh\left(\frac{x}{q} - r\right)$ will be equal to 1.

So $60 = p + 1 \Rightarrow p = 59$

$\Rightarrow y = 59 + \cosh\left(\frac{x}{q} - r\right)$

We have $\cosh\left(\frac{250}{q} - r\right) = 1$ at the minimum point, so $\frac{250}{q} - r = 0$

When $x = 0$, $y = 80$:

$80 = 59 + \cosh(-r)$

$\Rightarrow \cosh r = 21$

$\Rightarrow r = \operatorname{arcosh} 21 = \ln\left(21 + \sqrt{21^2 - 1}\right) = \ln\left(21 + 2\sqrt{110}\right) = 3.74$ (3 SF)

$\frac{250}{q} - \ln\left(21 + 2\sqrt{110}\right) = 0 \Rightarrow q = \frac{250}{\ln\left(21+2\sqrt{110}\right)} = 66.9$ (3 SF)

So $p = 59$, $q = 66.9$ (3 SF), $r = 3.74$ (3 SF)

7) b)

$y = 59 + \cosh\left(\frac{x}{66.897} - 3.7371\right)$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{66.897}\sinh\left(\frac{x}{66.897} - 3.7371\right)$

$s = \int_0^{500} \sqrt{1 + \left(\frac{1}{66.897}\sinh\left(\frac{x}{66.897} - 3.7371\right)\right)^2}\,\mathrm{d}x$

Note that this integral cannot be expressed in terms of elementary functions, so it must be evaluated numerically. In other words, we must use a calculator.

So $s = 503.225\ldots = 503$ metres to the nearest metre

8) a)

The minimum point of the curve is when $x = 1.6$, and the height will be 0 metres. At this point $\cosh(qx - r)$ will be equal to 1.

So $0 = p \times 1 - 2 \Rightarrow p = 2$

$\Rightarrow y = 2\cosh(qx - r) - 2$

We have $\cosh(1.6q - r) = 1$ at the minimum point, so $1.6q - r = 0$

When $x = 0$, $y = 1.5$:

$1.5 = 2\cosh(-r) - 2$

$\Rightarrow \cosh r = 1.75$

$\Rightarrow r = \operatorname{arcosh} 1.75 = \ln\left(1.75 + \sqrt{1.75^2 - 1}\right) = \ln\left(\frac{7}{4} + \frac{\sqrt{33}}{4}\right) = 1.15881 \ldots = 1.159$ (4 SF)

$1.6q - \ln\left(\frac{7}{4} + \frac{\sqrt{33}}{4}\right) = 0 \Rightarrow q = \frac{\ln\left(\frac{7}{4} + \frac{\sqrt{33}}{4}\right)}{1.6} = 0.724256 \ldots = 0.7243$ (4 SF)

So $p = 2$, $q = 0.7243$ (4 SF), $r = 1.158810 \ldots = 1.159$ (4 SF)

8) b)

$y = 2\cosh(0.72425 \ldots x - 1.1588 \ldots) - 2$

$y|_{x=3} = 2\cosh(1.01395906 \ldots) - 2 = 1.11927 \ldots$ metres

So CD is 1.12 metres tall to the nearest centimetre

8) c)

$y = 2\cosh(0.72425 \ldots x - 1.1588 \ldots) - 2$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 1.44851 \ldots \sinh(0.72425 \ldots x - 1.1588 \ldots)$

$s = \int_0^3 \sqrt{1 + (1.44851 \ldots \sinh(0.72425 \ldots x - 1.1588 \ldots))^2}\,\mathrm{d}x$

Note that this integral cannot be expressed in terms of elementary functions, so it must be evaluated numerically. In other words, we must use a calculator.

So $s = 4.1795 \ldots = 4.18$ metres to the nearest centimetre

9) a)

$y = \frac{1}{k}\cosh^2(kx)$ so $\frac{\mathrm{d}y}{\mathrm{d}x} = 2\cosh(kx)\sinh(kx) = \sinh(2kx)$

$s = \int_0^1 \sqrt{1 + (\sinh(2kx))^2}\,\mathrm{d}x = \int_0^1 \sqrt{\cosh^2(2kx)}\,\mathrm{d}x = \int_0^1 \cosh(2kx)\,\mathrm{d}x$

$= \left[\frac{1}{2k}\sinh(2kx)\right]_0^1 = \left(\frac{1}{2k}\sinh(2k)\right) - (0) = \frac{1}{2k}\sinh(2k)$ units

9) b)

$y = \frac{1}{k}\sinh^2(kx)$ so $\frac{\mathrm{d}y}{\mathrm{d}x} = 2\sinh(kx)\cosh(kx) = \sinh(2kx)$

$s = \int_0^1 \sqrt{1 + (\sinh(2kx))^2}\,\mathrm{d}x = \int_0^1 \sqrt{\cosh^2(2kx)}\,\mathrm{d}x = \int_0^1 \cosh(2kx)\,\mathrm{d}x$

$= \left[\frac{1}{2k}\sinh(2kx)\right]_0^1 = \left(\frac{1}{2k}\sinh(2k)\right) - (0) = \frac{1}{2k}\sinh(2k)$ units

10) a)

$x = \frac{1}{6}\sinh u$ so $\frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{6}\cosh u$ and $x^2 = \frac{1}{36}\sinh^2 u$

$\int \sqrt{1 + 36x^2}\,\mathrm{d}x = \int \frac{1}{6}\cosh u \sqrt{1 + 36\left(\frac{1}{36}\sinh^2 u\right)}\,\mathrm{d}u = \int \frac{1}{6}\sqrt{1 + \sinh^2 u}\,\mathrm{d}u =$

$\int \frac{1}{6}\cosh u \sqrt{\cosh^2 u}\,\mathrm{d}u = \int \frac{1}{6}\cosh^2 u\,\mathrm{d}u$

Since $\cosh 2A = 2\cosh^2 A - 1$ then $\cosh^2 A = \frac{1}{2} + \frac{1}{2}\cosh 2A$ so that

$\int \frac{1}{6}\cosh^2 u\,\mathrm{d}u = \int \frac{1}{6}\left(\frac{1}{2} + \frac{1}{2}\cosh 2u\right)\mathrm{d}u = \int \left(\frac{1}{12} + \frac{1}{12}\cosh 2u\right)\mathrm{d}u$

$= \frac{1}{12}u + \frac{1}{24}\sinh(2u) + c$

We have $u = \operatorname{arsinh}(6x)$ and

Since $\sinh(2u) = 2\sinh u \cosh u$ then $\sinh(2u) = 2(6x)\sqrt{1 + 36x^2} = 12x\sqrt{1 + 36x^2}$

$\int \sqrt{1 + 36x^2}\,\mathrm{d}x = \frac{1}{12}\operatorname{arsinh}(6x) + \frac{1}{2}x\sqrt{1 + 36x^2} + c$

10) b)

$y = 3x^2$ so $\frac{\mathrm{d}y}{\mathrm{d}x} = 6x$

$s = \int_0^{\frac{1}{3}} \sqrt{1 + (6x)^2}\mathrm{d}x = \int_0^{\frac{1}{3}} \sqrt{1 + 36x^2}\mathrm{d}x = \left[\frac{1}{12}\operatorname{arsinh}(6x) + \frac{1}{2}x\sqrt{1 + 36x^2}\right]_0^{\frac{1}{3}}$

$= \left(\frac{1}{12}\ln\left(2 + \sqrt{5}\right) + \frac{\sqrt{5}}{6}\right) - (0 + 0)$

$= \left(\frac{1}{12}\ln\left(2 + \sqrt{5}\right) + \frac{\sqrt{5}}{6}\right)$ units

11) a)

$x = \sinh^2 u$ so $\frac{\mathrm{d}x}{\mathrm{d}u} = 2\sinh u \cosh u$

$$\int \sqrt{\frac{x+1}{x}}\,\mathrm{d}x = \int \sqrt{\frac{\sinh^2 u + 1}{\sinh^2 u}}\,2\sinh u\cosh u\,\mathrm{d}u$$

$$= \int \sqrt{\frac{\cosh^2 u}{\sinh^2 u}}\,2\sinh u\cosh u\,\mathrm{d}u = \int \frac{\cosh u}{\sinh u}\,2\sinh u\cosh u\,\mathrm{d}u = \int 2\cosh^2 u\,\mathrm{d}u$$

Since $\cosh 2A = 2\cosh^2 A - 1$ then $\cosh^2 A = \frac{1}{2} + \frac{1}{2}\cosh 2A$ so that

$$\int 2\cosh^2 u\,\mathrm{d}u = \int (1 + \cosh 2u)\,\mathrm{d}u = u + \frac{1}{2}\sinh(2u) + c$$

Since $\sqrt{x} = \sinh u$ then $u = \operatorname{arsinh}\sqrt{x}$

Also $\cosh^2 u = 1 + \sinh^2 u$ so $\cosh u = \sqrt{x+1}$

This gives $\sinh 2u = 2\sinh u\cosh u = 2\sqrt{x}\sqrt{x+1} \Rightarrow \frac{1}{2}\sinh(2u) = \sqrt{x}\sqrt{x+1}$

$\therefore \int \sqrt{\frac{x+1}{x}}\,\mathrm{d}x = \operatorname{arsinh}\sqrt{x} + \sqrt{x}\sqrt{x+1} + c = \sqrt{x(x+1)} + \operatorname{arsinh}\sqrt{x} + c$ as req.

11) b)

$y = \frac{2}{3}x^{\frac{1}{2}}$ so $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{3}x^{-\frac{1}{2}}$

$$s = \int_1^4 \sqrt{1 + \left(\frac{1}{3}x^{-\frac{1}{2}}\right)^2}\,\mathrm{d}x = \int_1^4 \sqrt{1 + \frac{1}{9x}}\,\mathrm{d}x = \int_1^4 \sqrt{\frac{9x+1}{9x}}\,\mathrm{d}x$$

Make the substitution $u = 9x$ so $\frac{\mathrm{d}u}{\mathrm{d}x} = 9$

When $x = 1$, $u = 9$ and when $x = 4$, $u = 36$

$$\int_1^4 \sqrt{\frac{9x+1}{9x}}\,\mathrm{d}x = \int_9^{36} \frac{1}{9}\sqrt{\frac{u+1}{u}}\,\mathrm{d}u = \frac{1}{9}\left[\operatorname{arsinh}\sqrt{u} + \sqrt{u}\sqrt{u+1}\right]_9^{36}$$

$$= \frac{1}{9}\left[\left(\ln\left(6+\sqrt{37}\right) + 6\sqrt{37}\right) - \left(\ln\left(3+\sqrt{10}\right) + 3\sqrt{10}\right)\right]$$

$$= \left(\frac{1}{9}\ln\left(\frac{6+\sqrt{37}}{3+\sqrt{10}}\right) + \frac{2\sqrt{37}}{3} - \frac{\sqrt{10}}{3}\right) \text{ units}$$

12) a)

$y = 5 - \ln(1 - x^2)$ so $\frac{dy}{dx} = \frac{2x}{1-x^2}$

$1 + \left(\frac{2x}{1-x^2}\right)^2 = \frac{(1-x^2)^2}{(1-x^2)^2} + \left(\frac{2x}{1-x^2}\right)^2 = \frac{1-2x^2+x^4}{(1-x^2)^2} + \frac{4x^2}{(1-x^2)^2} = \frac{1+2x^2+x^4}{(1-x^2)^2} = \frac{(1+x^2)^2}{(1-x^2)^2}$

$\therefore \int \sqrt{1 + \left(\frac{2x}{1-x^2}\right)^2}\, dx = \int \sqrt{\frac{(1+x^2)^2}{(1-x^2)^2}}\, dx = \int \left(\frac{1+x^2}{1-x^2}\right) dx$

12) b)

$\frac{2x^2}{1-x^2} = A + \frac{B}{1-x} + \frac{C}{1+x}$

$1 + x^2 = A(1-x)(1+x) + B(1+x) + C(1-x)$

Let $x = 1$: $\quad 2 = 2B \therefore B = 1$

Let $x = -1$: $\quad 2 = 2C \therefore C = 1$

Let $x = 0$: $\quad 1 = A + B + C \therefore A = -1$

$\therefore \frac{2x^2}{1-x^2} = -1 + \frac{1}{1-x} + \frac{1}{1+x}$

$\int_0^{\frac{1}{2}} \left(\frac{1+x^2}{1-x^2}\right) dx = \int_0^{\frac{1}{2}} \left(-1 + \frac{1}{1-x} + \frac{1}{1+x}\right) dx$

$= [-x - \ln(1-x) + \ln(1+x)]_0^{\frac{1}{2}}$

$= \left(-\frac{1}{2} - \ln\frac{1}{2} + \ln\frac{3}{2}\right) - (-0 - 0 + 0)$

$= \left(-\frac{1}{2} + \ln 3\right)$ units

13)

$y = \frac{1}{3}x^{\frac{3}{2}}$ so $\frac{dy}{dx} = \frac{1}{2}x^{\frac{1}{2}}$

$\int_0^k \sqrt{1 + \left(\frac{1}{2}x^{\frac{1}{2}}\right)^2}\, dx = \int_0^k \sqrt{1 + \frac{1}{4}x}\, dx = \frac{1}{2}\int_0^k \sqrt{4+x}\, dx = \frac{1}{2}\int_0^k (4+x)^{\frac{1}{2}}\, dx$

$= \left[\frac{1}{3}(4+x)^{\frac{3}{2}}\right]_0^k = \left(\frac{1}{3}(4+k)^{\frac{3}{2}}\right) - \left(\frac{8}{3}\right)$

So $\frac{1}{3}(4+k)^{\frac{3}{2}} - \frac{8}{3} = 39$

$\Rightarrow \frac{1}{3}(4+k)^{\frac{3}{2}} = \frac{125}{3} \Rightarrow (4+k)^{\frac{3}{2}} = 125 \Rightarrow 4 + k = 25 \therefore x = 21$

14) a)

At $x = 30$, there is a minimum height so that $\frac{30}{b} - c = 0$

And $12 = a + \cosh(0) = a + 1 \therefore a = 11$

At $(0, 20)$, $20 = 11 + \cosh(-c) = 11 + \cosh(c)$

$\Rightarrow 9 = \cosh c$

$\therefore c = \operatorname{arcosh} 9 = \ln\left(9 + \sqrt{80}\right)$

So that $\frac{30}{b} - \ln\left(9 + \sqrt{80}\right) = 0 \therefore b = \frac{30}{\ln\left(9+\sqrt{80}\right)}$

$\therefore a = 11$, $b = \frac{30}{\ln\left(9+\sqrt{80}\right)}$ and $c = \ln\left(9 + \sqrt{80}\right)$

14) b)

$h = a + \cosh\left(\frac{x}{b} - c\right)$ so $\frac{\mathrm{d}h}{\mathrm{d}x} = \frac{1}{b}\sinh\left(\frac{x}{b} - c\right)$

For $b = \frac{30}{\ln\left(9+\sqrt{80}\right)}$ and $c = \ln\left(9 + \sqrt{80}\right)$

$\int_0^{60} \sqrt{1 + \frac{1}{b^2}\sinh^2\left(\frac{x}{b} - c\right)}\,\mathrm{d}x = 63.448027\ldots$

$\therefore 63$ metres (to the nearest metre)

15) a)

$\frac{\mathrm{d}x}{\mathrm{d}t} = 2t \Rightarrow \left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 = 4t^2$

$\frac{\mathrm{d}y}{\mathrm{d}t} = t^2 - 1 \Rightarrow \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2 = (t^2 - 1)^2 = t^2 - 2t^2 + 1$

$\therefore \left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2 = 4t^2 + t^4 - 2t^2 + 1 = t^4 + 2t^2 + 1 = (t^2 + 1)^2$ as req.

15) b)

$s = \int_0^4 \sqrt{(t^2 + 1)^2}\mathrm{d}t = \int_0^4 (t^2 + 1)\mathrm{d}t = \left[\frac{1}{3}t^3 + t\right]_0^4 = \left(\frac{64}{3} + 4\right) - (0 + 0) = \frac{76}{3}$ units

16) a)

$\frac{dx}{dt} = 2t^2 \Rightarrow \left(\frac{dx}{dt}\right)^2 = (2t^2)^2 = 4t^4$

$\frac{dy}{dt} = 4\sqrt{2}t \Rightarrow \left(\frac{dy}{dt}\right)^2 = \left(4\sqrt{2}t\right)^2 = 32t^2$

$\therefore \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 4t^4 + 32t^2$

$s = \int_0^2 \sqrt{4t^4 + 32t^2}\,dt = \int_0^2 \sqrt{4t^2(t^2+8)}\,dt = \int_0^2 2t\sqrt{t^2+8}\,dt$ as req.

16) b)

$\int_0^2 2t\sqrt{t^2+8}\,dt = \int_0^2 2t(t^2+8)^{\frac{1}{2}}\,dt$

$= \left[\frac{2}{3}(t^2+8)^{\frac{3}{2}}\right]_0^2 = \frac{2}{3} \times 12^{\frac{3}{2}} - \frac{2}{3} \times 8^{\frac{3}{2}} = \left(16\sqrt{3} - \frac{32}{3}\sqrt{2}\right)$ units

17)

$\frac{dx}{dt} = \frac{5}{2}(1+t)^{-\frac{1}{2}} \Rightarrow \left(\frac{dx}{dt}\right)^2 = \left(\frac{5}{2}(1+t)^{-\frac{1}{2}}\right)^2 = \frac{25}{4(1+t)}$

$\frac{dy}{dt} = -\frac{5}{2}(1-t)^{-\frac{1}{2}} \Rightarrow \left(\frac{dy}{dt}\right)^2 = \left(-\frac{5}{2}(1-t)^{-\frac{1}{2}}\right)^2 = \frac{25}{4(1-t)}$

$\therefore \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = \frac{25}{4(1+t)} + \frac{25}{4(1-t)} = \frac{25(1-t)+25(1+t)}{4(1+t)(1-t)} = \frac{25}{2(1-t^2)}$

When $x = 5$, $t = 0$ and when $x = 5\sqrt{2}$, $t = 1$

$s = \int_0^1 \sqrt{\frac{25}{2(1-t^2)}}\,dt = \int_0^1 \frac{5\sqrt{2}}{2} \times \frac{1}{\sqrt{1-t^2}}\,dt = \left[\frac{5\sqrt{2}}{2}\arcsin(t)\right]_0^1 = \frac{5\sqrt{2}}{4}\pi - 0 = \frac{5\sqrt{2}}{4}\pi$ units

18)

$\frac{dx}{d\theta} = -r\sin\theta \Rightarrow \left(\frac{dx}{d\theta}\right)^2 = r^2\sin^2\theta$

$\frac{dy}{d\theta} = r\cos\theta \Rightarrow \left(\frac{dy}{d\theta}\right)^2 = r^2\cos^2\theta$

$\therefore \left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 = r^2\sin^2\theta + r^2\cos^2\theta = r^2(\sin^2\theta + \cos^2\theta) = r^2$

$s = \int_0^{2\pi} \sqrt{r^2}\,d\theta = \int_0^{2\pi} r\,d\theta = [r\theta]_0^{2\pi} = (2\pi r) - (0) = 2\pi r$

19)

$\frac{dx}{dt} = e^{-t}(\cos t - \sin t) - e^{-t}(\cos t + \sin t) = e^{-t}(\cos t - \sin t - \cos t - \sin t)$

$= e^{-t}(-2\sin t)$

$\Rightarrow \left(\frac{dx}{dt}\right)^2 = 4e^{-2t}\sin^2 t$

$\frac{dy}{dt} = e^{-t}(-\sin t - \cos t) - e^{-t}(\cos t - \sin t) = e^{-t}(-\sin t - \cos t - \cos t + \sin t)$

$= e^{-t}(-2\cos t)$

$\Rightarrow \left(\frac{dy}{dt}\right)^2 = 4e^{-2t}\cos^2 t$

$\therefore \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 4e^{-2t}\sin^2 t + 4e^{-2t}\cos^2 t = 4e^{-2t}(\sin^2 t + \cos^2 t) = 4e^{-2t}$

$s = \int_0^3 \sqrt{4e^{-2t}}dt = \int_0^3 2e^{-t}dt = [-2e^{-t}]_0^3 = -2e^{-3} + 2 = \left(-\frac{2}{e^3} + 2\right)$ units

20) a)

$\frac{dx}{dt} = -3\sin t \Rightarrow \left(\frac{dx}{dt}\right)^2 = 9\sin^2 t$

$\frac{dy}{dt} = 3 + 3\cos t \Rightarrow \left(\frac{dy}{dt}\right)^2 = (3 + 3\cos t)^2 = 9 + 18\cos t + 9\cos^2 t$

$\therefore \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 9\sin^2 t + 9 + 18\cos t + 9\cos^2 t = 18 + 18\cos t = 18(1 + \cos t)$

Since $\cos 2A = \cos^2 A - \sin^2 A$ then

$18(1 + \cos t) = 18\left(1 + \cos^2\left(\frac{t}{2}\right) - \sin^2\left(\frac{t}{2}\right)\right) = 18\left(\cos^2\left(\frac{t}{2}\right) + \cos^2\left(\frac{t}{2}\right)\right)$

$= 36\cos^2\left(\frac{t}{2}\right)$ as req.

20) b)

$s = \int_0^{\pi} \sqrt{36\cos^2\left(\frac{t}{2}\right)}\,dt = \int_0^{\pi} 6\cos\left(\frac{t}{2}\right)dt = \left[12\sin\left(\frac{t}{2}\right)\right]_0^{\pi} = (12) - (0) = 12$ units

21)

$\frac{\mathrm{d}x}{\mathrm{d}t} = 9\sin^2 t\cos t \Rightarrow \left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 = (9\sin^2 t\cos t)^2 = 81\sin^4 t\cos^2 t$

$\frac{\mathrm{d}y}{\mathrm{d}t} = -9\cos^2 t\sin t \Rightarrow \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2 = (-9\cos^2 t\sin t)^2 = 81\cos^4 t\sin^2 t$

$\therefore \left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2$

$= 81\sin^4 t\cos^2 t + 81\cos^4 t\sin^2 t$

$= 81\sin^2 t\cos^2 t\,(\sin^2 t + \cos^2 t)$

$= 81\sin^2 t\cos^2 t$

$= \frac{81}{4}\sin^2(2t)$

$s = \int_0^{\frac{\pi}{2}} \sqrt{\frac{81}{4}\sin^2(2t)}\,\mathrm{d}t = \int_0^{\frac{\pi}{2}} \frac{9}{2}\sin(2t)\,\mathrm{d}t = \left[-\frac{9}{4}\cos(2t)\right]_0^{\frac{\pi}{2}} = \left(\frac{9}{4}\right) - \left(-\frac{9}{4}\right) = \frac{9}{2}$ units

22) a)

$\frac{\mathrm{d}x}{\mathrm{d}t} = 4\cos t - 4\cos 4t$

$\Rightarrow \left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 = (4\cos t - 4\cos 4t)^2 = 16\cos^2 t - 32\cos 4t\cos t + 16\cos^2 4t$

$\frac{\mathrm{d}y}{\mathrm{d}t} = -4\sin t + 4\sin 4t$

$\Rightarrow \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2 = (-4\sin t + 4\sin 4t)^2 = 16\sin^2 t - 32\sin 4t\sin t + 16\sin^2 4t$

$\therefore \left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2$

$= 16\cos^2 t - 32\cos 4t\cos t + 16\cos^2 4t + 16\sin^2 t - 32\sin 4t\sin t + 16\sin^2 4t$

$= 16(\cos^2 t + \sin^2 t) - 32(\cos 4t\cos t + \sin 4t\sin t) + 16(\cos^2 4t + \sin^2 4t)$

$= 16 - 32\cos(4t - t) + 16$

$= 32 - 32\cos(3t)$

$= 64\left(\frac{1}{2} - \frac{1}{2}\cos(3t)\right)$

$= 64\sin^2\left(\frac{3t}{2}\right)$ as req.

22) b)

$s = \int_0^{\frac{\pi}{3}} \sqrt{64\sin^2\left(\frac{3t}{2}\right)}\,\mathrm{d}t = \int_0^{\frac{\pi}{3}} 8\sin\left(\frac{3t}{2}\right)\mathrm{d}t = \left[-\frac{16}{3}\cos\left(\frac{3t}{2}\right)\right]_0^{\frac{\pi}{3}} = (0) - \left(-\frac{16}{3}\right) = \frac{16}{3}$ units

23)

$\frac{dx}{dt} = k(1+\cos t) \Rightarrow \left(\frac{dx}{dt}\right)^2 = k^2(1+\cos t)^2 = k^2(1+2\cos t+\cos^2 t)$

$\frac{dy}{dt} = k\sin t \Rightarrow \left(\frac{dy}{dt}\right)^2 = k^2(\sin t)^2 = k^2\sin^2 t$

$\therefore \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = k^2(1+2\cos t+\cos^2 t) + k^2\sin^2 t$

$= k^2(1+2\cos t+\cos^2 t+\sin^2 t)$

$= k^2(2+2\cos t)$

$= 4k^2\cos^2\left(\frac{t}{2}\right)$

$s = \int_0^{\pi}\sqrt{4k^2\cos^2\left(\frac{t}{2}\right)}\,dt = \int_0^{\pi} 2k\cos\left(\frac{t}{2}\right)dt = \left[4k\cos\left(\frac{t}{2}\right)\right]_0^{\pi} = (4k)-(0) = 4k$

But $s = 12$

$\therefore k = 3$

24) a)

$\int \operatorname{sech} x\,dx = \int \frac{1}{\cosh x}dx = \int \frac{1}{\cosh x}dx$

Let $u = \sinh x$ so $\frac{du}{dx} = \cosh x$ and $1+u^2 = 1+\sinh^2 x = \cosh^2 x$ so that

$\int \frac{1}{\cosh x}dx = \int \frac{\cosh x}{\cosh^2 x}dx = \int \frac{1}{1+u^2}du = \arctan u + c = \arctan(\sinh x) + c$ as req.

24) b)

$\frac{dx}{dt} = -\frac{\sinh t}{\cosh^2 t} = -\tanh t\operatorname{sech} t \Rightarrow \left(\frac{dx}{dt}\right)^2 = (-\tanh t\operatorname{sech} t)^2 = \tanh^2 t\operatorname{sech}^2 t$

$\frac{dy}{dt} = \operatorname{sech}^2 t \Rightarrow \left(\frac{dy}{dt}\right)^2 = (\operatorname{sech}^2 t)^2 = \operatorname{sech}^4 t$

$\therefore \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2$

$= \tanh^2 t\operatorname{sech}^2 t + \operatorname{sech}^4 t$

$= \operatorname{sech}^2 t\,(\tanh^2 t + \operatorname{sech}^2 t)$

$= \operatorname{sech}^2 t$

$\int_0^{\ln 2}\sqrt{\operatorname{sech}^2 t}\,dt = \int_0^{\ln 2}\operatorname{sech} t\,dt = [\arctan(\sinh t)]_0^{\ln 2} = \left(\arctan\left(\frac{3}{4}\right)\right) - (0)$

$= \arctan\left(\frac{3}{4}\right)$ units

25) a)

$\int \sqrt{x^2+1}\,\mathrm{d}x$

Let $x = \sinh u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \cosh u \Rightarrow \mathrm{d}x = \cosh u\,\mathrm{d}u$

$\therefore \int \sqrt{x^2+1}\,\mathrm{d}x$

$= \int \sqrt{\sinh^2 u - 1} \times \cosh u\,\mathrm{d}u$

$= \int \sqrt{\cosh^2 u} \times \cosh u\,\mathrm{d}u$

$= \int \cosh u \times \cosh u\,\mathrm{d}u$

$= \int \cosh^2 u\,\mathrm{d}u$

$\cosh 2u \equiv 2\cosh^2 u - 1 \Rightarrow \cosh^2 u \equiv \frac{1}{2}(\cosh 2u + 1)$

$\therefore \int \cosh^2 u\,\mathrm{d}u = \int \left(\frac{1}{2}(\cosh 2u + 1)\right)\mathrm{d}u$

$= \frac{1}{2}\int (\cosh 2u + 1)\,\mathrm{d}u$

$= \frac{1}{2}\left(\frac{1}{2}\sinh 2u + u\right) + c$

$= \frac{1}{4}\sinh 2u + \frac{1}{2}u + c$

$\sinh 2u = 2\sinh u \cosh u = 2\sinh u \times \sqrt{\sinh^2 u + 1} = 2x\sqrt{x^2+1}$

$\therefore \int \sqrt{x^2+1}\,\mathrm{d}x = \frac{1}{4} \times 2x\sqrt{x^2+1} + \frac{1}{2}\operatorname{arsinh} x + c$

$= \frac{1}{2}x\sqrt{x^2+1} + \frac{1}{2}\operatorname{arsinh} x + c$

$= \frac{1}{2}x\sqrt{x^2+1} + \frac{1}{2}\ln\left(x + \sqrt{x^2+1}\right) + c$ as req.

25) b)

$\frac{\mathrm{d}x}{\mathrm{d}t} = 2at \Rightarrow \left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 = (2at)^2 = 4a^2t^2$

$\frac{\mathrm{d}y}{\mathrm{d}t} = 4a \Rightarrow \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2 = (4a)^2 = 16a^2$

$\therefore \left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2 = 4a^2t^2 + 16a^2 = 4a^2(t^2 + 4)$

$\int_0^1 \sqrt{4a^2(t^2+4)}\mathrm{d}t = \int_0^1 2a\sqrt{4+t^2}\mathrm{d}t = \int_0^1 4a\sqrt{1+\left(\frac{t}{2}\right)^2}\,\mathrm{d}t$

Let $u = \frac{t}{2}$ so $\frac{\mathrm{d}u}{\mathrm{d}t} = \frac{1}{2}$ when $t = 0, u = 0$, when $t = 1$, $u = \frac{1}{2}$

$\int_0^1 4a\sqrt{1+\left(\frac{t}{2}\right)^2}\,\mathrm{d}t = \int_0^{\frac{1}{2}} 8a\sqrt{1+u^2}\mathrm{d}t$

$= 8a\left[\frac{1}{2}u\sqrt{1+u^2} + \frac{1}{2}\ln\left(u+\sqrt{1+u^2}\right)\right]_0^{\frac{1}{2}} = 8a\left(\frac{\sqrt{5}}{8} + \frac{1}{2}\ln\left(\frac{1}{2}+\frac{\sqrt{5}}{2}\right)\right) - (0)$

$= 4a\left(\frac{\sqrt{5}}{4} + \ln\left(\frac{1}{2}+\frac{\sqrt{5}}{2}\right)\right)$ units